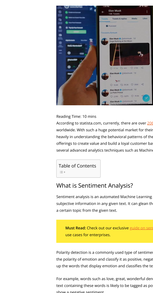

Twitter Sentiment Analysis: What is it and How to Perform [Real Example]

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Gramener
Insights as Stories
Twitter Sentiment Analysis

Reading Time: 10 mins

According to statista.com, currently, there are over 206 million monetizable daily users of Twitter worldwide. With such a huge potential market for their products and services, companies are sure to invest heavily in understanding the behavioral patterns of their key demographics so that they can tailor their offerings to create value and build a loyal customer base. Twitter sentiment analysis of customers using several advanced analytics techniques such as Machine Learning or Text analysis is a way to do that.

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What is Sentiment Analysis?

Sentiment analysis is an automated Machine Learning process that helps identify, classify, and analyze subjective information in any given text. It can glean the emotions, opinions, or judgments of people about a certain topic from the given text.

Must Read: Check out our exclusive guide on sentiment analysis to understand its benefits and use cases for enterprises.

Polarity detection is a commonly used type of sentiment analysis. Through this technique, one can identify the polarity of emotion and classify it as positive, negative, or neutral. This sentiment analysis model picks up the words that display emotion and classifies the text accordingly.

For example, words such as love, great, wonderful denote a positive sentiment most of the time and hence, text containing these words is likely to be tagged as positive. Similarly, words such as hate, dislike, grudge show a negative sentiment.

Sentiment analysis uses Natural Language Processing (NLP) and Machine Learning to come up with the results. Businesses can use sentiment analysis as a tool to stay up-to-date about the sentiments around their business or brand.

What is Twitter Sentiment Analysis?

Let's say that you are the customer experience manager of Acme Corp, a product-based company that's selling computer monitors across 125 countries. Your customers use Twitter to share their feedback on your product. Some appreciate your product, while some criticize it for its shape, key sizes, or something else.

As a CX manager, you want to know the sentiment of customers on Twitter. You can develop an **Automated Machine Learning Sentiment Analysis Model** to scrape the Twitter data and compute the customer perception.

Twitter sentiment analysis, which is a subset of social media sentiment analysis, helps firms to understand their audience on social channels, stay on top of what's being said about their brand – and their rivals – and uncover new trends in the market by carefully listening to the voice of the consumer on Twitter.

On Twitter, there are a bazillion discussions going on — talks that have no boundaries. Politicians use Twitter to speak their agendas and progress with the audience. Businesses and brands use Twitter to communicate with their customers in real-time.

Unfortunately, the amount of data generated is so voluminous that apart from user data, there is a large amount of non-useful data, often known as noise. This is why it's difficult for businesses to know which tweets to respond to initially. Twitter sentiment analysis fills this void.

The Automated Machine Learning Sentiment Analysis Model has been developed to understand customer perception from the data collected from Twitter. This is why sentiment analysis has become a crucial tool for social media marketing strategies.

Also Check Out: A Comprehensive Guide on Customer Sentiment Analysis to know the problems it solves for enterprises.

What Insights Can Twitter Sentiment Analysis Provide?

Twitter sentiment analysis can help make sense of huge amounts of data in real-time. Doing so manually would take thousands of man-hours and may still be suspect to human bias and not give accurate results.

Social media monitoring can not only just help businesses, it can also help politicians, sports teams, volunteer organizations, and more. Let's have a look at some of the use cases of Twitter sentiment analysis.

Political and Public Opinions

Twitter sentiment analysis has tremendous use for government and political leaders. It helps them stay abreast of the public opinion about their parties, their actions, and their statements.

One wrong statement can sway the public opinion negatively on Twitter and, in an election season, this may prove to be detrimental.

Similarly, sentiment analysis can help public servants understand how government policies and actions have affected the public psyche. Newsrooms do Twitter analysis to understand the sentiments of citizens during elections. Predicting the election results based on public opinion on Twitter is a common use case.

Sports

Twitter sentiment analysis data can be very helpful in understanding public support for teams which can boost or suppress team morale. For example, If team India is performing fantastically in cricket, the public sentiment around the team and its players is usually positive.

This is reflected in the tweets and hashtags used by people. Similarly, sports directors may find it useful to analyze the tweets of their teams' players to understand their mental and emotional state. Tweets may reveal feelings such as anger, sadness, fear, or happiness, thereby, helping the sports directors know and manage their teams better.

Identifying Social Evils

Twitter sentiment analysis around specific social topics such as bullying, harassment, molestation, dowry, rape, etc. can guide government and NGO actions in preventing the same.

Understanding what people are saying about such social evils can help form policies to combat such social evils. Talking about such topics in a face-to-face setting is often taboo and tough for people. This is why Twitter and other social media platforms become outlets for venting strong emotions.

Hate Speech Identification

The automated Machine Learning process can identify words that denote hate speech. There can be hate speech against politicians, celebrities, ideologies, etc. By knowing who and where the hate speech is coming from, remedial actions can be taken.

Understanding Customer Sentiments for Business

Businesses and brands can monitor what their customers are saying. Analyzing customer sentiment pre-launch and post-launch of products can help businesses modify their products as per customer need.

Businesses can also understand the gaps in customer needs and create new products or modify old ones. Aspect-based sentiment analysis of the Twitter feed can help businesses gain insight about which aspects they need to work on and what gives them a competitive edge over their rivals.

How to do Twitter Sentiment Analysis?

Common steps to perform Twitter sentiment analysis include:

1. Sort Twitter data
2. Clean Twitter data
3. Develop Twitter sentiment analysis model
4. Analyze Twitter data for positive/negative sentiments
5. Visualize the insights

Sort Twitter Data

The first step in sentiment analysis is the collection and sorting of data. There is a sea of data on Twitter, it is important to pick the data that is most relevant to the problem you are looking to solve or the thing you wish to find out. Only relevant data can be used to train the sentiment analysis model and test whether the model performs satisfactorily on Twitter data. Another important aspect to cover is what type of tweets you are looking to analyze – historical or current. To sort this data, you first need to extract it from Twitter. For this, you can use some of the following platforms:

- Zapier to create an automated workflow between Twitter and Google Sheets, for example.
- IFTTT to collect Twitter data without any code.
- Export Tweet to track hashtags, keywords, etc. in real-time or look for historical tweets and mentions.
- Tweet Download to collect tweets from one's own account including the mentions and replies.
- Twitter API to access and analyze public tweets about keywords, brand mentions, hashtags, or tweets from particular people.
- Tweepy, the python library, for reaching Twitter API and gathering data from there.

Twitter Data Cleansing

Once the data is gathered and sorted, it then needs to be cleaned before it can be used to train the Twitter sentiment analysis model. Twitter data is mostly unstructured, so the cleaning process involves removing emojis, special characters, and unnecessary blank spaces. The process also includes getting rid of duplicate tweets, making format adjustments, and also removing very short tweets – those less than three characters. Clean data can give more accurate results.

Develop Twitter Sentiment Analysis Model

There are different Machine Learning platforms that can help one build and implement a Twitter sentiment analysis model. These platforms can provide access to pre-trained or ready-to-train models. You can use your Twitter data to train these models. For developing a model, you need to go through the following steps:

- Choose the type of model you want to build, for example, a classifier model that helps classify text into pre-defined tags.
- Decide the type of classification. In this case, it would be Sentiment Analysis.
- Import the relevant Twitter data to train the model.
- Tag the data as positive, negative, or neutral, for example, to train your model.
- Test your model.

Analyze Twitter Data for Sentiments

Once your model is trained and gives satisfactory test results, it is ready for deployment. Now, you simply need to connect your Twitter data with your sentiment analysis model. There are several ways to accomplish it. One way to do it is to analyze a particular file of new or unseen tweets and classify them. Another way is to integrate Twitter data with Zapier and Google sheets and analyze this data using your model.

Insight Visualization

There are tools that help visualize your data results and make them easy to interpret and digest. These attractive visualization tools such as Google Data Studio, Looker, Tableau, etc. create visual reports including charts, graphs, and data tables that are easily understood by a larger number of people.

ChatGPT Enhancing Twitter Sentiment Analysis

Social media platforms like Twitter serve as a goldmine of public opinion. Analyzing the sentiment behind tweets not only provides insights into individual reactions but also paints a broader picture of public perception. With the advent of advanced AI models like **ChatGPT**, Twitter sentiment analysis has reached new heights. Here's how ChatGPT is making a significant impact:

- Contextual Understanding: Twitter is a hub of concise, context-heavy conversations. ChatGPT excels in understanding the intricacies of language, including slang, idioms, and abbreviations commonly used on Twitter. Its ability to grasp contextual nuances ensures accurate sentiment interpretation, even in the midst of brief and cryptic tweets.
- **Emotion Decoding: Emojis**, GIFs, and emoticons are integral to Twitter communication. ChatGPT can decipher the emotional tone conveyed through these symbols, adding an extra layer of depth to sentiment analysis. Understanding not just the words but also the emotions expressed enhances the overall accuracy of the analysis.
- **Real-Time Engagement:** Twitter operates in real-time, with trends and topics changing swiftly. ChatGPT's capacity for instant responses enables businesses and researchers to engage with Twitter users dynamically. By interacting with users in real-time, ChatGPT gathers live feedback, enriching the sentiment analysis process with fresh, up-to-the-minute data.
- **Language Adaptability:** Twitter is a global platform where tweets are posted in numerous languages. ChatGPT's multilingual capabilities allow it to handle tweets in various languages, ensuring a comprehensive analysis regardless of the linguistic diversity of the tweets being examined.
- **Continuous Learning:** ChatGPT is continually learning from the vast array of conversations it encounters. This ongoing learning process enables it to adapt to evolving language trends, slang, and emerging terminologies on Twitter. Consequently, its analysis remains relevant and reflective of the current linguistic landscape.

By incorporating ChatGPT into Twitter sentiment analysis, businesses, researchers, and analysts gain access to a tool that not only understands the language intricacies of tweets but also decodes the underlying emotions. This deeper understanding not only improves the accuracy of sentiment analysis but also provides actionable insights for informed decision-making.

Check out our latest blog on ChatGPT for Sentiment Analysis and learn how the Large Language Models (LLMs) are augmenting customer experience.

Benefits of Twitter Sentiment Analysis

Twitter sentiment analysis can help you control all mentions around your brand from a single place. You can monitor, research, and react to billions of conversations around your business using sentiment analysis. Following are the key benefits for users of Twitter data sentiment analysis:

Discover Brand Perception

The Twitter sentiment analysis dataset can give you a bird' eye view of your brand perception. You can know what people are saying about your brand and your clients. Knowing the brand perception can help you identify potential problems as well as reap benefits from hidden opportunities. Sometimes, just a single mention from a reputed blogger can catapult your brand to fame and give a fillip to your business. Some of the other benefits of monitoring your brand include:

- Using brand monitoring tools can help you identify your brand ambassadors – the people who have a real connection with the masses.
- Anticipating Pre-identifying trends can help you become a pioneer in your field. You can target customers with promotions and information that is trending the most.
- By knowing the threats to your brand or business, you can engage in quick firefighting to salvage your brand reputation and averting a looming brand crisis.
- Competitor brand monitoring can provide you useful insights for business tactics and brand communication ideas that are working for them.

Grow Your Influence

Improving brand perception by using social media brand ambassadors can help you grow your influence among your audience. Twitter sentiment analysis tool can help you identify these brand ambassadors and connect with them. You can also create social media strategies that are in sync with the sentiment analysis and help you gain more popularity. You can improve your brand's influence by quickly responding to both the positive and negative sentiments of people.

Improve Customer Service

In a world where everything is available at the click of a button, customers want instant solutions to their problems also at the click of a button. This is why it is important to track customer problems using the sentiment analysis tool on Twitter and employ customer service agents who can instantly resolve their problems. While it would be impossible for these agents to manually sift through a sea of data to find tweets that need an immediate response, the Twitter sentiment analysis tool does that for them.

Gramener's Data Stories on Twitter Data

Gramener has developed an application that offers a holistic understanding of the feelings and emotions associated with keywords on Twitter. Users can run search queries against which the application will pull out the latest 100-200 tweets containing the keyword and categorize them based on the positive or negative sentiments expressed in the tweets.

The original tweets, retweets, and replies are color-coded to distinguish them from each other. Light-colored bubbles are the actual tweets, orange bubbles are the retweets, and the blue bubbles represent the replies.

The bubble size is proportional to the size of the following the tweeter enjoys. The bubbles are mapped along the X and Y-axis. The x-axis represents the age of the tweets, oldest to newest. The Y-axis represents the sentiment.

In addition to the sentiment analysis of the tweets containing the keywords, the application also creates a word cloud using the terms common to the tweets. Clicking on any of the words will instantly display the sentiment analysis of the tweets that contain the word.

The Twitter trending application has been built on the Gramex platform. Using the Twitter API it can fetch around 2000 tweets at a time. It performs the Sentiment Analysis using a Python Library called Textblob.

Conclusion

There is no doubt that Sentiment Analysis can provide an enviable edge to entrepreneurs trying to get ahead of the curve. With more people now shopping online than before because of the pandemic and sharing their customer experiences on social media, it behooves brands to closely listen to what their target audience is saying about their products and services.

A report from Market Research Future claims that the global Sentiment Analytics market is expected to grow to USD 6 billion by 2027. With its ability to deliver invaluable insights about customer behavior and target audience persona, Sentiment Analysis is sure to be a part of the Twitter marketing strategy of corporates for the foreseeable future.

Contact us for custom built low code data and AI solutions for your business challenges and check out unstructured data analytics solutions built for our clients, including Fortune 500 companies. Book a free demo right now.

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